

18.1 Electric Fields

Question Paper

Course	CIEA Level Physics
Section	18. Electric Fields
Topic	18.1 Electric Fields
Difficulty	Medium

Time allowed: 30

Score: /23

Percentage: /100

Question 1

State Coulomb's law.

[2 marks]

Question 2a

State what is represented by an electric field line.

[2 marks]

Question 2b

Two point charges A and B are placed 0.014 mm apart as shown in Fig. 1.1.

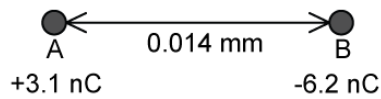


Fig. 1.1

The charge of A is +3.1 nC and the charge of B is -6.2 nC. The charges have the following radii:

$$r_A = 3.2 \text{ pm}$$

$$r_B = 4.1 \text{ pm}$$

Determine the magnitude of the force between the two charges.

[3 marks]

Question 2c

The magnitude of the electric forces between the charges is very high.

State and explain why this is the case.

[2 marks]

Question 2d

(i)
Draw arrows on Fig. 1.1 to show the direction of the electrostatic force coming from charges A and B.

[2]

(ii)
Particle B is replaced by particle C. Particle C has a charge of $+6.2 \text{ nC}$.

Describe how the electrostatic force between charges A and C will be different to that between A and B.

[2]

(iii)
A positron is placed at point Y, equidistant from both A and B, as shown in Fig. 1.2.

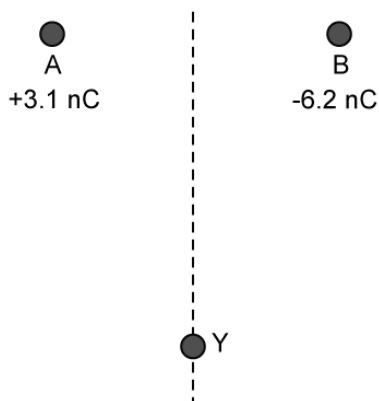


Fig. 1.2

On Fig. 1.2 draw an arrow representing the direction of the resultant force acting on the positron.

[1]

[5 marks]

Question 3a

State what is indicated by the direction of the electric field line.

[2 marks]

Question 3b

Fig. 1.1 shows a pair of parallel metal plates with a potential difference (p.d.) of 3500 V between them.

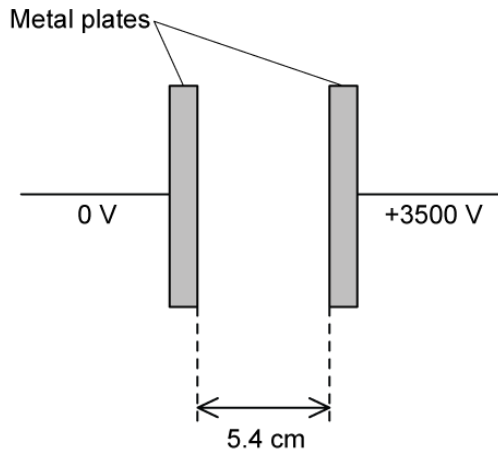


Fig. 1.1

The plates are separated by a distance of 4.6 cm. The plates are in a vacuum.

On Fig. 1.1, draw five lines to represent the electric field in the region between the plates.

[3 marks]

Question 3c

Calculate the strength of the electric field between the plates.

electric field strength = N C^{-1}

[2 marks]

Question 3d

A moving electron enters the region between the plates from the left, as shown in Fig. 1.2.

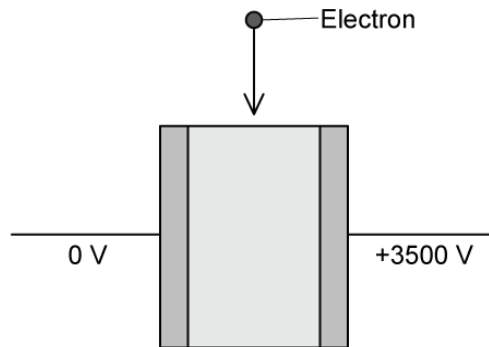


Fig. 1.2

The electron is deflected by the electric field.

On Fig. 1.2, draw a line to show the path of the electron as it moves through and out of the region of the electric field.

[2 marks]

Model Answers: Medium

1

State Coulomb's law:

- The electrostatic force between two point charges; [1 mark]
- Is proportional to the product of the charges and inversely proportional to the square of their separation; [1 mark]

[Total: 2 marks]

2a

(a) The electric field line represents:

- The direction of the (electric) force; [1 mark]
- The force is on a positive charge; [1 mark]

[Total: 2 marks]

2b

(b) Determine the magnitude of the force between the two charges:

List the known quantities:

- Charge A, $Q_A = +3.1 \text{ nC} = 3.1 \times 10^{-9} \text{ C}$
- Charge B, $Q_B = -6.2 \text{ nC} = -6.2 \times 10^{-9} \text{ C}$
- Radius of charge A, $r_A = 3.2 \text{ pm} = 3.2 \times 10^{-12} \text{ m}$
- Radius of charge B, $r_B = 4.1 \text{ pm} = 4.1 \times 10^{-12} \text{ m}$
- Distance between charges, $R = 0.014 \text{ mm} = 0.014 \times 10^{-3} \text{ m}$
- $\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ m F}^{-1}$

Calculate the distance between the charges, r :

- $r = r_A + r_B + R$

- $r = (3.2 \times 10^{-12}) + (4.1 \times 10^{-12}) + (0.014 \times 10^{-3})$
- $r = 1.4 \times 10^{-5} \text{ m}$ [1 mark]

Use Coulomb's Law to calculate the force between the charges:

- $F = \frac{Q_A Q_B}{4 \pi \epsilon_0 r^2}$
- $F = (8.99 \times 10^9) \times \frac{(3.1 \times 10^{-9}) \times (6.2 \times 10^{-9})}{(1.4 \times 10^{-5})^2}$ [1 mark]
- $F = 881.6 \text{ N} = 880 \text{ N (2 s.f.)}$ [1 mark]

[Total: 3 marks]

The question is asking for the **magnitude** of the force, hence the **magnitude** of the charges only should be used in the calculation.

2c

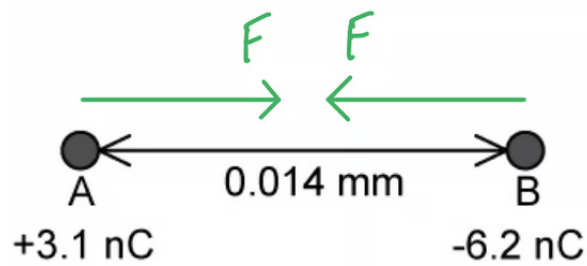
- (c) The magnitude of the electric force between the charges is very high because:
- The charges are very close together / the separation of the charges is small / r is small; [1 mark]
 - Electric force follows an inverse square law; [1 mark]

[Total: 2 marks]

2d

- (d)
- (i) Draw arrows on Fig. 1.1 to show the direction of the electrostatic force coming from charges A and B

A diagram that would score 2 marks should look like this:



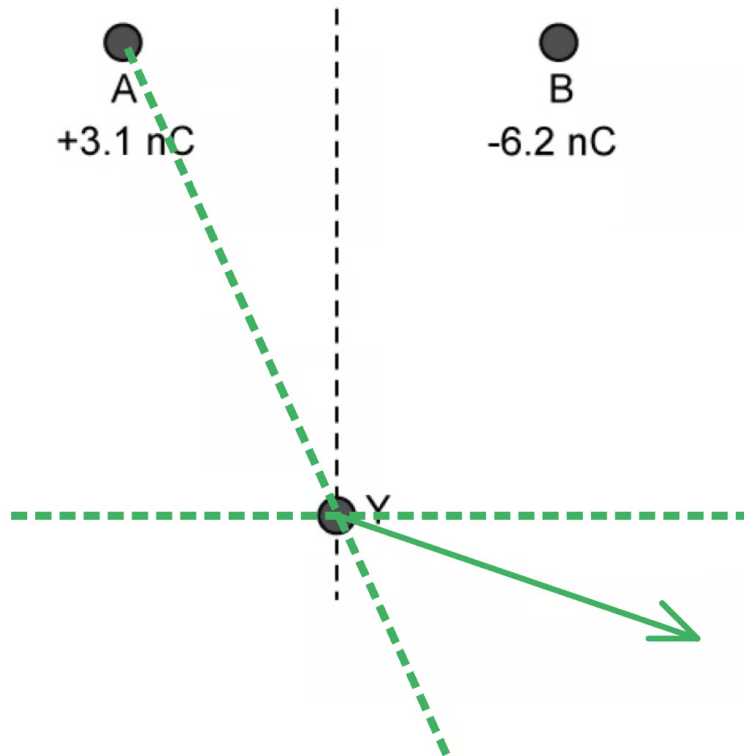
- Two arrows of equal length labelled F / force; [1 mark]
- Facing towards each other; [1 mark]

(ii) The electrostatic force between charges A and C will be different to that between A and B because:

- Both charges A and C are positive, so there is a repulsive force between them; [1 mark]
- Hence, the electrostatic force will be directed away from both particles; [1 mark]

(iii) On Fig. 1.2 draw an arrow representing the direction of the resultant force acting on the positron:

A diagram that would score 1 mark should look like this:



- A straight arrow drawn to the right of Y at an angle; [1 mark]

[Total: 5 marks]

3a

(a) The direction of the electric field lines indicates:

- The direction of the (electrostatic) force; [1 mark]
- The force on the positive charge; [1 mark]

[Total: 2 marks]

3b

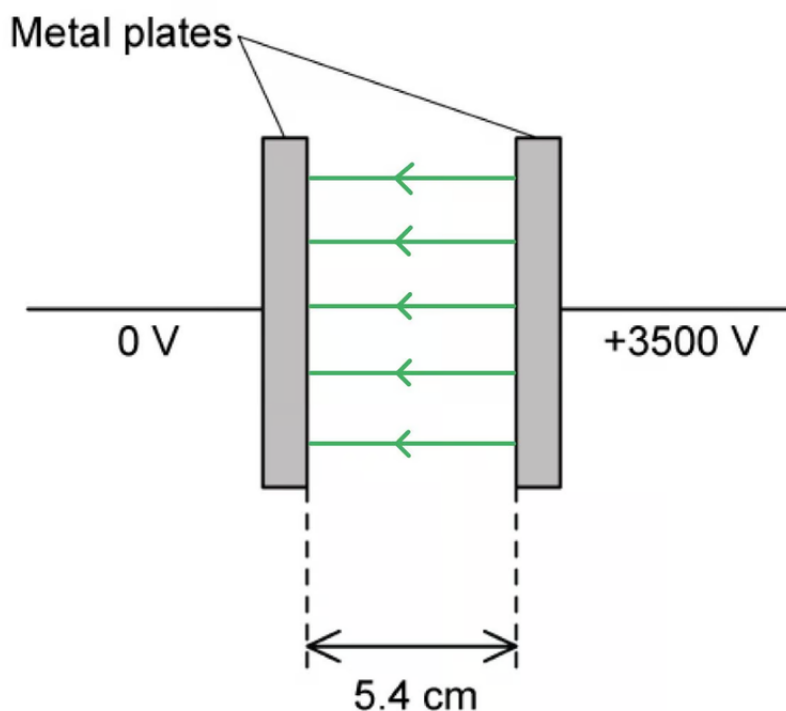
(b) On Fig. 1.1, draw five lines to represent the electric field in the region between the plates:

A diagram that would score 3 marks should look like this:

3b

(b) On Fig. 1.1, draw five lines to represent the electric field in the region between the plates:

A diagram that would score 3 marks should look like this:



- One straight line is drawn perpendicular to the two plates, starting on one plate and finishing on the other; [1 mark]
- Five straight lines are drawn parallel to the one above, uniformly spaced; [1 mark]
- Arrows are drawn on each line from the right side to the left side plate; [1 mark]

[Total: 3 marks]

3c

(c) Calculate the strength of the electric field between the plates:

List the known quantities:

- Potential difference between the plates, $V = 3500 \text{ V}$
- Separation between the plates, $d = 5.4 \text{ cm} = 0.054 \text{ m}$

Use the equation for electric field strength:

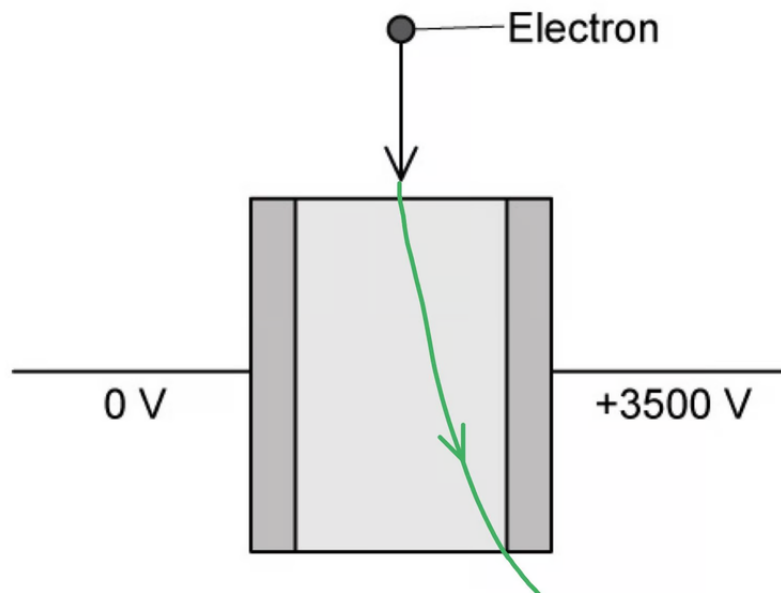
- $E = \frac{V}{d}$ [1 mark]
- $E = \frac{3500}{0.054}$
- $E = 6.48 \times 10^4 \text{ N C}^{-1} = 6.5 \times 10^4 \text{ N C}^{-1} \text{ (2 s.f.)}$ [1 mark]

[Total: 2 marks]

3d

(d) On Fig. 1.2, draw a line to show the path of the electron as it moves through and out of the region of the electric field:

A diagram that would score 2 marks should look like this:



- Electron path is straight outside the field and a smooth curve inside; [1 mark]
- The direction of deflection is shown as towards the positive plate; [1 mark]

[Total: 2 marks]